



A Level • Cambridge (CIE) • Chemistry

36 mins

19 questions

Multiple Choice Practice

Topical Practice MCQ

9. The Periodic Table: chemical periodicity

Carefully selected past paper questions by topic.

16 Which statement is correct?

- A** Aluminium chloride has a giant ionic lattice of Al^{3+} and Cl^{-} ions.
- B** Sodium chloride dissolves in water, forming hydrogen chloride and sodium hydroxide.
- C** The strong covalent bonds in silicon chloride prevent it from reacting with water.
- D** When phosphorus(V) chloride is added to water, the resulting solution conducts electricity.

- 17 Elements X and Y are in Period 3 of the Periodic Table. Element X is either phosphorus or sulfur. Element Y is either sodium or magnesium.

Element X forms an oxide that reacts with water to give a solution containing the aqueous anion XO_4^{2-} .

One mole of element Y reacts with one mole of chlorine molecules. At the end of the reaction, all of the element Y and all of the chlorine molecules have been used up.

What are elements X and Y?

	X	Y
A	phosphorus	sodium
B	phosphorus	magnesium
C	sulfur	sodium
D	sulfur	magnesium

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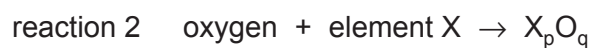
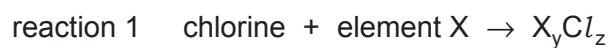
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C	sulfur	sodium
D	sulfur	magnesium

18 In reactions 1 and 2, X represents an element in Period 3.

In each reaction, X is forming a product where X is in its highest oxidation state.



Which ratios show a steady increase from sodium to phosphorus?

- A neither $z:y$ nor $q:p$
- B $z:y$ only
- C $q:p$ only
- D both $z:y$ and $q:p$

16 X, Y and Z are elements all found within Groups 13, 14 and 15 of the Periodic Table.

X is in the same group in the Periodic Table as Y.

Y and Z are in Period 3.

The first ionisation energy of X is greater than the first ionisation energy of Y.

The melting point of Z is less than the melting point of Y.

Y and Z both form chlorides which are white solids. These white solids react with water to produce solutions with a pH of less than 4.

Which row of the table shows the possible identities of X and Y?

	X	Y
A	B	Al
B	Ge	Si
C	As	P
D	N	P

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- 17 $\text{NH}_3(\text{aq})$ is added to separate samples of $\text{NaCl}(\text{aq})$, $\text{MgCl}_2(\text{aq})$, $\text{BaCl}_2(\text{aq})$ and $\text{SiCl}_4(\text{l})$. Under the conditions of this experiment, only two samples will produce a white precipitate when $\text{NH}_3(\text{aq})$ is added.

What are these two samples?

- A $\text{MgCl}_2(\text{aq})$ and $\text{BaCl}_2(\text{aq})$
- B $\text{MgCl}_2(\text{aq})$ and $\text{SiCl}_4(\text{l})$
- C $\text{NaCl}(\text{aq})$ and $\text{BaCl}_2(\text{aq})$
- D $\text{NaCl}(\text{aq})$ and $\text{SiCl}_4(\text{l})$

- 25** Separate 1.0 g samples of Na_2O , MgO , Al_2O_3 , SiO_2 , NaCl , MgCl_2 , Al_2Cl_6 and SiCl_4 are added to separate beakers containing water and stirred.

The number of beakers containing a white solid is Q.

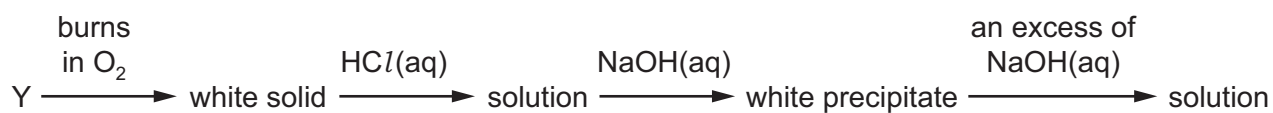
An excess of NaOH(aq) is then added to each beaker and stirred.

The number of beakers now containing a white solid is R.

Which row is correct?

	Q	R
A	3	2
B	3	3
C	4	3
D	4	4

13 An element, Y, reacts according to the following sequence.



What could be element Y?

A Na

B Mg

C Al

D P

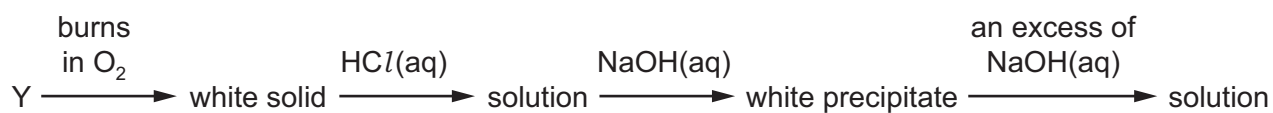
- 12 X, Y and Z are elements in Period 3 of the Periodic Table. The results of some experiments carried out with compounds of these elements are shown.

element	result of adding the oxide of the element to $\text{H}_2\text{O}(\text{l})$	result of adding the chloride of the element to $\text{H}_2\text{O}(\text{l})$	result of adding the oxide of the element to $\text{HCl}(\text{aq})$
X	no reaction	hydrolyses	forms chloride salt
Y	forms hydroxide	dissolves	forms chloride salt
Z	forms acid	hydrolyses	hydrolyses

Which statement could be correct?

- A X is Al and Y is Mg.
- B X is Si and Y is Na.
- C Y is Al and Z is P.
- D Y is Na and Z is Al.

13 An element Y reacts according to the following sequence.



What could be element Y?

A Al

B Ca

C Mg

D P

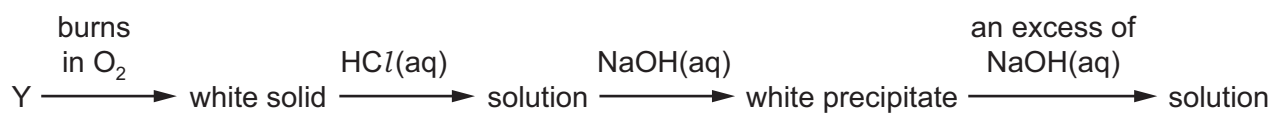
13 Element Z has a giant structure.

The chloride of Z reacts with water to give a solution with a pH less than 5.

Which row shows two elements which could be Z?

- A** aluminium, magnesium
- B** aluminium, silicon
- C** phosphorus, magnesium
- D** phosphorus, silicon

13 An element Y reacts according to the following sequence.



What could be element Y?

A Al

B Ca

C Mg

D P

- 12** The oxide and chloride of an element **X** are separately mixed with water. The two resulting solutions have the same effect on litmus.

What could element **X** be?

A *Al*

B Ca

C Na

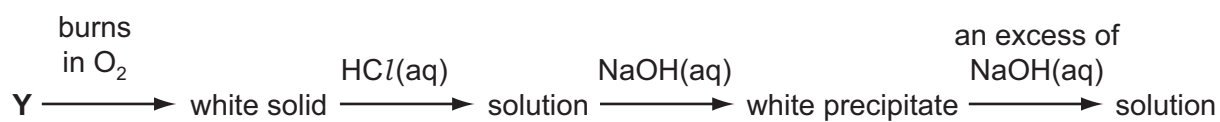
D P

- 15 The labels had become detached from four bottles in the laboratory. A student realised that the contents of one of them could easily be identified, because on addition of water it would **not** give fumes of hydrogen chloride.

Which did **not** give the HCl fumes?

- A** Al_2Cl_6 **B** MgCl_2 **C** PCl_5 **D** SiCl_4

17 An element **Y** reacts according to the following sequence.



What could be element **Y**?

A Al

B Ca

C Mg

D P

Question 17

9701_w11_qp_11 • Q1

- 1 Three elements, **X**, **Y** and **Z**, have the physical properties shown in the table.

element	melting point / °C	boiling point / °C	density / g cm ⁻³
X	-7	59	3.12
Y	98	883	0.97
Z	649	1107	1.74

What could be the identities of **X**, **Y** and **Z**?

	X	Y	Z
A	Br ₂	Al	Si
B	Br ₂	Na	Mg
C	I ₂	Mg	Na
D	I ₂	Si	K

- 13** Which statement correctly describes what happens when silicon tetrachloride is added to water?
- A** The SiCl_4 dissolves to give a neutral solution only.
 - B** The SiCl_4 reacts to give an acidic solution only.
 - C** The SiCl_4 reacts to give a precipitate and an acidic solution.
 - D** The SiCl_4 reacts to give a precipitate and a neutral solution.

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- A The SiCl_4 dissolves to give a neutral solution only.
 - B The SiCl_4 reacts to give an acidic solution only.
 - C The SiCl_4 reacts to give a precipitate and an acidic solution.
 - D The SiCl_4 reacts to give a precipitate and a neutral solution.

Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_12	9. The Periodic Table: chemical per...	D
2	9701_w24_qp_11	9. The Periodic Table: chemical per...	D
3	9701_w24_qp_13	9. The Periodic Table: chemical per...	D
4	9701_s23_qp_12	9. The Periodic Table: chemical per...	D
5	9701_w23_qp_11	9. The Periodic Table: chemical per...	A
6	9701_w23_qp_13	9. The Periodic Table: chemical per...	A
7	9701_s22_qp_12	9. The Periodic Table: chemical per...	B
8	9701_w22_qp_13	9. The Periodic Table: chemical per...	D
9	9701_w21_qp_11	9. The Periodic Table: chemical per...	C
10	9701_m19_qp_12	9. The Periodic Table: chemical per...	A
11	9701_w17_qp_11	9. The Periodic Table: chemical per...	A
12	9701_w17_qp_12	9. The Periodic Table: chemical per...	B
13	9701_w17_qp_13	9. The Periodic Table: chemical per...	A
14	9701_s16_qp_13	9. The Periodic Table: chemical per...	D
15	9701_s15_qp_13	9. The Periodic Table: chemical per...	B
16	9701_w14_qp_13	9. The Periodic Table: chemical per...	A
17	9701_w11_qp_11	9. The Periodic Table: chemical per...	B
18	9701_w09_qp_11	9. The Periodic Table: chemical per...	C
19	9701_w09_qp_12	9. The Periodic Table: chemical per...	C